```python
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns

df = pd.read_csv('mymoviedb.csv', lineterminator='\n')

df.head()
```

```
  Release_Date                    Title  \
0   2021-12-15  Spider-Man: No Way Home
1   2022-03-01               The Batman
2   2022-02-25                  No Exit
3   2021-11-24                  Encanto
4   2021-12-22           The King's Man

                                            Overview  Popularity
Vote_Count  \
0  Peter Parker is unmasked and no longer able to...    5083.954
8940
1  In his second year of fighting crime, Batman u...    3827.658
1151
2  Stranded at a rest stop in the mountains durin...    2618.087
122
3  The tale of an extraordinary family, the Madri...    2402.201
5076
4  As a collection of history's worst tyrants and...    1895.511
1793

   Vote_Average Original_Language                               Genre
\
0           8.3                en  Action, Adventure, Science Fiction

1           8.1                en            Crime, Mystery, Thriller

2           6.3                en                            Thriller

3           7.7                en  Animation, Comedy, Family, Fantasy

4           7.0                en    Action, Adventure, Thriller, War


                                          Poster_Url
0  https://image.tmdb.org/t/p/original/1g0dhYtq4i...
1  https://image.tmdb.org/t/p/original/74xTEgt7R3...
2  https://image.tmdb.org/t/p/original/vDHsLnOWKl...
3  https://image.tmdb.org/t/p/original/4j0PNHkMr5...
4  https://image.tmdb.org/t/p/original/aq4Pwv5Xeu...
```

```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Release_Date       9827 non-null   str
 1   Title              9827 non-null   str
 2   Overview           9827 non-null   str
 3   Popularity         9827 non-null   float64
 4   Vote_Count         9827 non-null   int64
 5   Vote_Average       9827 non-null   float64
 6   Original_Language  9827 non-null   str
 7   Genre              9827 non-null   str
 8   Poster_Url         9827 non-null   str
dtypes: float64(2), int64(1), str(6)
memory usage: 691.1 KB
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

exploration summary

dataframe consist 9827 rows and 9 column we do not have duplicated values release date column needs to casted into date time and extract only year value Overview , Original_Language and Poster_Url no need so drop them there is noticable outliers in Popularity column genre column have comma and white space so casted them

```
df['Release_Date'] = pd.to_datetime(df['Release_Date'])
```

```
print(df['Release_Date'].dtypes)
```

```
datetime64[us]
```

```
df['Release_Date'] = pd.to_datetime(df['Release_Date'])
```

```
df['Release_Date'] = df['Release_Date'].dt.year
```

```
df['Release_Date'].dtypes
```

```
dtype('int32')
```

```
df.head()
```

```
   Release_Date                    Title  \
0          1970  Spider-Man: No Way Home
1          1970  Spider-Man: No Way Home
2          1970  Spider-Man: No Way Home
3          1970               The Batman
4          1970               The Batman
```

```
                                            Overview  Popularity  
Vote_Count  \
0  Peter Parker is unmasked and no longer able to...    5083.954   
8940   
1  Peter Parker is unmasked and no longer able to...    5083.954   
8940   
2  Peter Parker is unmasked and no longer able to...    5083.954   
8940   
3  In his second year of fighting crime, Batman u...    3827.658   
1151   
4  In his second year of fighting crime, Batman u...    3827.658   
1151   

   Vote_Average Original_Language            Genre  \
0           8.3                en           Action   
1           8.3                en        Adventure   
2           8.3                en  Science Fiction   
3           8.1                en            Crime   
4           8.1                en          Mystery   

                                          Poster_Url  
0  https://image.tmdb.org/t/p/original/1g0dhYtq4i...  
1  https://image.tmdb.org/t/p/original/1g0dhYtq4i...  
2  https://image.tmdb.org/t/p/original/1g0dhYtq4i...  
3  https://image.tmdb.org/t/p/original/74xTEgt7R3...  
4  https://image.tmdb.org/t/p/original/74xTEgt7R3...  
```

DROPING THE COLUMNS

```
cols = ['Overview','Original_Language','Poster_Url']

df.drop(cols , axis =1, inplace = True)
df.columns
```

```
Index(['Release_Date', 'Title', 'Popularity', 'Vote_Count',
'Vote_Average',
       'Genre'],
      dtype='str')
```

```
df.head()
```

```
   Release_Date                     Title  Popularity  Vote_Count  \
0          2021  Spider-Man: No Way Home    5083.954        8940   
1          2022                The Batman    3827.658        1151   
2          2022                   No Exit    2618.087         122   
3          2021                   Encanto    2402.201        5076   
4          2021            The King's Man    1895.511        1793   

   Vote_Average                                  Genre  
0           8.3  Action, Adventure, Science Fiction  
```

```
1           8.1            Crime, Mystery, Thriller
2           6.3                            Thriller
3           7.7  Animation, Comedy, Family, Fantasy
4           7.0    Action, Adventure, Thriller, War
```

**categories the Vote avg column**

popular , average, below agerage, not popular

```
def categorise_col(df,col,labels):
    edges = [df[col].describe() ['min'],
            df[col].describe() ['25%'],
            df[col].describe() ['50%'],
            df[col].describe() ['75%'],
            df[col].describe() ['max']]
    df[col]= pd.cut(df[col],edges , labels = labels, duplicates =
'drop')
    return df


df = pd.read_csv('mymoviedb.csv', lineterminator='\n')

cols = ['Overview','Original_Language','Poster_Url']

df.drop(cols , axis =1, inplace = True)
df.columns

Index(['Release_Date', 'Title', 'Popularity', 'Vote_Count',
'Vote_Average',
       'Genre'],
      dtype='str')

df.head()

  Release_Date                    Title  Popularity  Vote_Count
Vote_Average  \
0   2021-12-15  Spider-Man: No Way Home    5083.954        8940
8.3
1   2022-03-01               The Batman    3827.658        1151
8.1
2   2022-02-25                  No Exit    2618.087         122
6.3
3   2021-11-24                  Encanto    2402.201        5076
7.7
4   2021-12-22           The King's Man    1895.511        1793
7.0

                                Genre
0  Action, Adventure, Science Fiction
1            Crime, Mystery, Thriller
2                            Thriller
```

```
3  Animation, Comedy, Family, Fantasy
4    Action, Adventure, Thriller, War
```

```
labels = ['not popular' , 'below average', 'average', 'popular']
categorise_col(df,'Vote_Average',labels)
```

```
df['Vote_Average'].unique()
```

```
['popular', 'below average', 'average', 'not popular', NaN]
Categories (4, str): ['not popular' < 'below average' < 'average' <
'popular']
```

```
df['Vote_Average'].value_counts()
```

```
Vote_Average
not popular      2467
popular          2450
average          2412
below average    2398
Name: count, dtype: int64
```

```
df.dropna(inplace = True)
```

```
df.isna().sum()
```

```
Release_Date    0
Title           0
Popularity      0
Vote_Count      0
Vote_Average    0
Genre           0
dtype: int64
```

```
df.head()
```

```
  Release_Date                    Title  Popularity  Vote_Count  \
0   2021-12-15  Spider-Man: No Way Home    5083.954        8940
1   2022-03-01               The Batman    3827.658        1151
2   2022-02-25                  No Exit    2618.087         122
3   2021-11-24                  Encanto    2402.201        5076
4   2021-12-22           The King's Man    1895.511        1793

    Vote_Average                               Genre
0        popular  Action, Adventure, Science Fiction
1        popular            Crime, Mystery, Thriller
2  below average                            Thriller
3        popular  Animation, Comedy, Family, Fantasy
4        average    Action, Adventure, Thriller, War
```

**now split genre into a list and and dataframe have only one genre for each row**

```python
df['Genre'] = df['Genre'].str.split(', ')

df = df.explode('Genre').reset_index(drop=True)

df.head()
```

```
  Release_Date                    Title  Popularity  Vote_Count
Vote_Average  \
0   2021-12-15  Spider-Man: No Way Home    5083.954        8940
popular
1   2021-12-15  Spider-Man: No Way Home    5083.954        8940
popular
2   2021-12-15  Spider-Man: No Way Home    5083.954        8940
popular
3   2022-03-01               The Batman    3827.658        1151
popular
4   2022-03-01               The Batman    3827.658        1151
popular

             Genre
0           Action
1        Adventure
2  Science Fiction
3            Crime
4          Mystery
```

```python
## casting column into category

df['Genre'] = df['Genre'].astype('category')
df['Genre'].dtypes
```

```
CategoricalDtype(categories=['Action', 'Adventure', 'Animation',
'Comedy', 'Crime',
                  'Documentary', 'Drama', 'Family', 'Fantasy',
'History',
                  'Horror', 'Music', 'Mystery', 'Romance', 'Science
Fiction',
                  'TV Movie', 'Thriller', 'War', 'Western'],
, ordered=False, categories_dtype=str)
```

```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 25552 entries, 0 to 25551
Data columns (total 6 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Release_Date  25552 non-null  str
 1   Title         25552 non-null  str
 2   Popularity    25552 non-null  float64
 3   Vote_Count    25552 non-null  int64
```

```
 4   Vote_Average  25552 non-null  category
 5   Genre         25552 non-null  category
dtypes: category(2), float64(1), int64(1), str(2)
memory usage: 848.9 KB
```

```
df.nunique()
```

```
Release_Date     100
Title           9415
Popularity      8088
Vote_Count      3265
Vote_Average       4
Genre             19
dtype: int64
```

#**DATA VISUALIZATION**

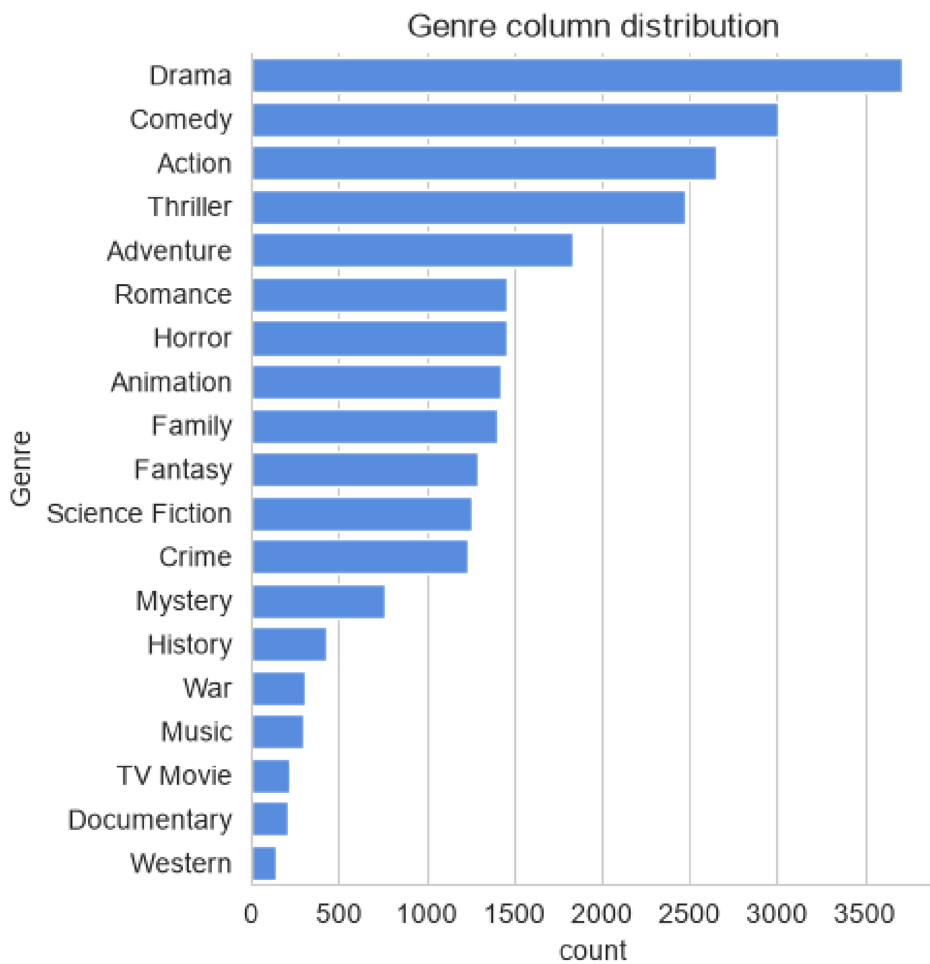
```
sns.set_style('whitegrid')
```

**what is most frequent genre of movies released on netflex?**

```
df['Genre'].describe()
```

```
count     25552
unique       19
top       Drama
freq       3715
Name: Genre, dtype: object
```

```
sns.catplot(y = 'Genre', data = df, kind = 'count',
            order = df['Genre'].value_counts().index,
            color = '#4287f5')
plt.title('Genre column distribution')
plt.show()
```

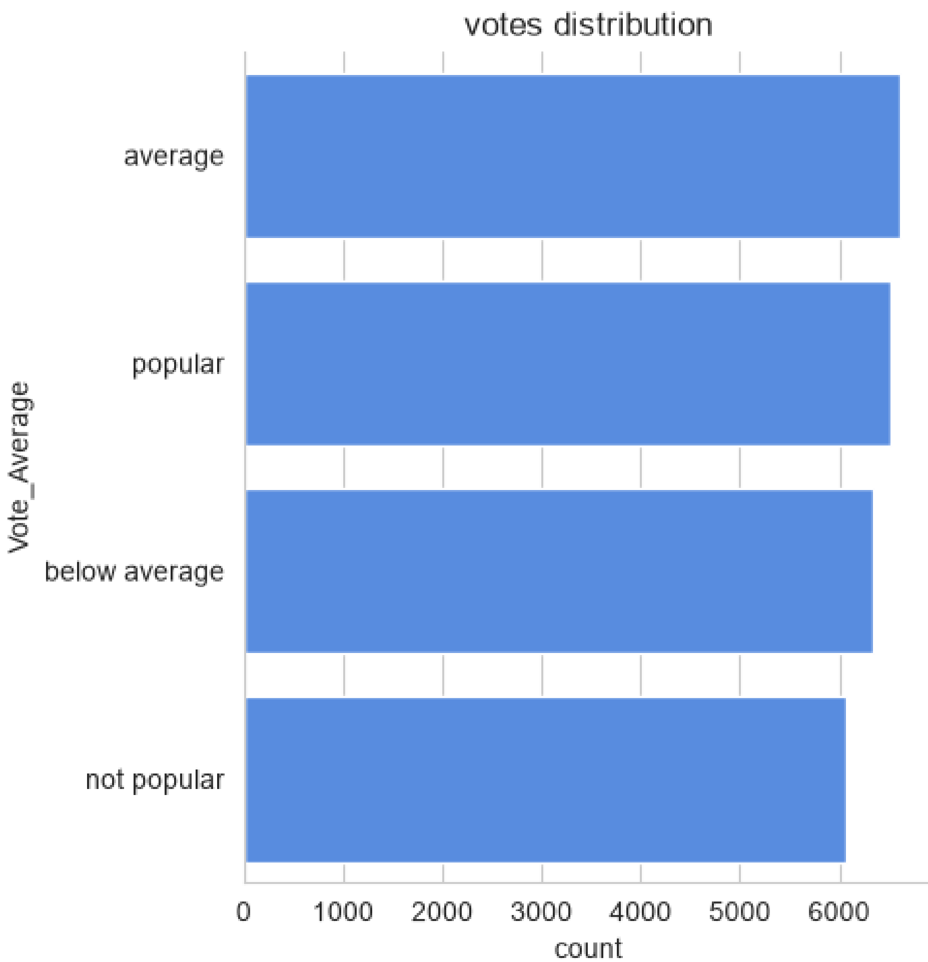



# which has highest votes in vote column ?

```
df.head()

   Release_Date                    Title  Popularity  Vote_Count
Vote_Average  \
0         2021  Spider-Man: No Way Home    5083.954        8940
popular
1         2021  Spider-Man: No Way Home    5083.954        8940
popular
2         2021  Spider-Man: No Way Home    5083.954        8940
popular
3         2022               The Batman    3827.658        1151
popular
4         2022               The Batman    3827.658        1151
popular

            Genre
0          Action
```

```
1         Adventure
2   Science Fiction
3             Crime
4           Mystery
```

```
sns.catplot(y = 'Vote_Average', data = df, kind = 'count',
            order = df['Vote_Average'].value_counts().index,
            color = '#4287f5')
plt.title('votes distribution')
plt.show()
```




#which movie gots the highest popularity and whats its genre?

```
df[df['Popularity'] == df['Popularity'].max()]

   Release_Date                    Title  Popularity  Vote_Count
Vote_Average  \
0          2021  Spider-Man: No Way Home    5083.954        8940
popular
1          2021  Spider-Man: No Way Home    5083.954        8940
```

```
popular
2         2021  Spider-Man: No Way Home    5083.954        8940
popular

             Genre
0           Action
1        Adventure
2  Science Fiction
```

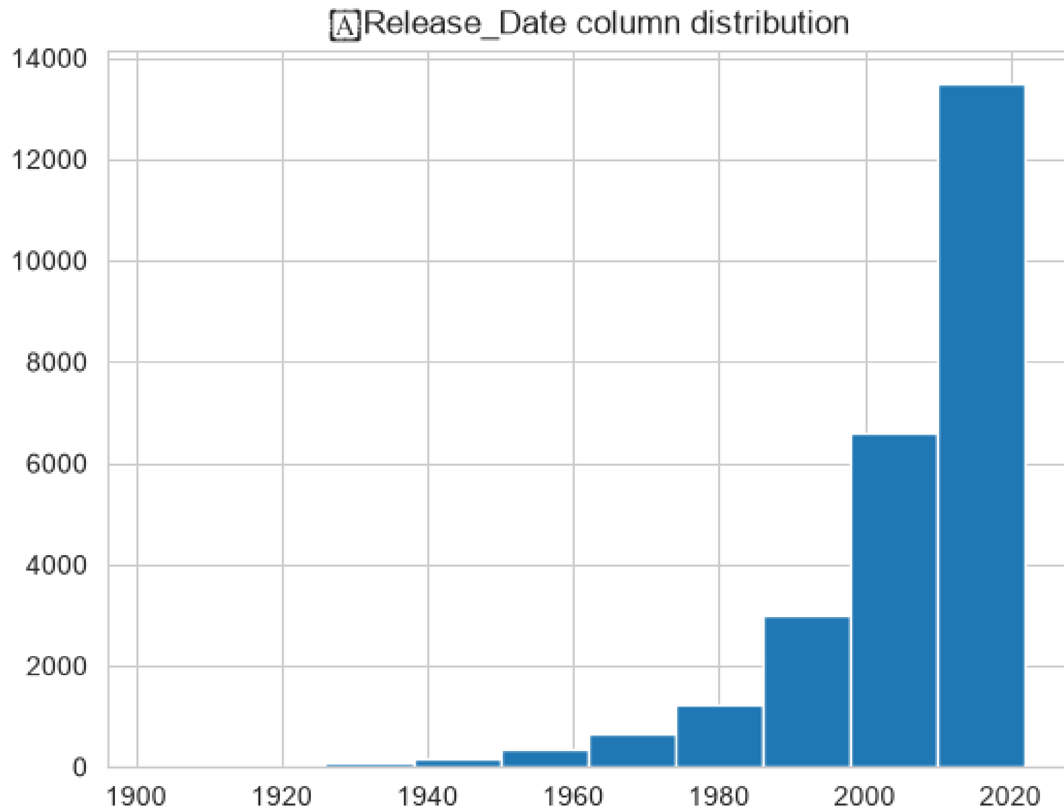
#which movie gots the less popularity and whats its genre?

```
df[df['Popularity'] == df['Popularity'].min()]
```

```
      Release_Date                                 Title  Popularity  \
25546         2021  The United States vs. Billie Holiday      13.354

25547         2021  The United States vs. Billie Holiday      13.354

25548         2021  The United States vs. Billie Holiday      13.354

25549         1984                               Threads      13.354

25550         1984                               Threads      13.354

25551         1984                               Threads      13.354


      Vote_Count Vote_Average            Genre
25546        152      average            Music
25547        152      average            Drama
25548        152      average          History
25549        186      popular              War
25550        186      popular            Drama
25551        186      popular  Science Fiction
```

#which year has most filmmed movies?

```
df['Release_Date'].hist()
plt.title('     Release_Date column distribution')
plt.show()

C:\Users\rbcho\AppData\Roaming\Python\Python311\site-packages\IPython\
core\pylabtools.py:170: UserWarning: Glyph 9 (   ) missing from
font(s) Arial.
  fig.canvas.print_figure(bytes_io, **kw)
```


Release_Date column distribution
14000
12000
10000
8000
6000
4000
2000
0
1900
1920
1940
1960
1980
2000
2020